

Explanatory Summarization with Discourse-Driven Planning

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Introduction

Lay Summarization Challenge:

- Scientific texts need **explanatory content** for accessibility
- Current methods **underrepresent** explanations in generated summaries

Key Insights:

- Most models follow end-to-end approaches **without explicit explanation modeling**
- Plan-based models offer better **controllability** and reduce hallucinations

The cerebellum utilizes proprioceptive feedback to fine-tune the timing of movements in a sequence based on previous actions.

q1: How does the cerebellum use feedback to adjust the timing of movements in a sequence?

Imagine the cerebellum as a coach who watches how you perform a move, then gives tips to improve the next one based on what was seen.

Figure 1. An example of plan question

Method

Step 1: Explanatory Content Extraction

- Apply **DMRST** parser to identify explanatory EDUs from RST relations
- Focus on 4 key relations: *Background*, *Elaboration*, *Explanation*, *Comparison*
- Extract (explanation, target) EDU pairs from reference summaries

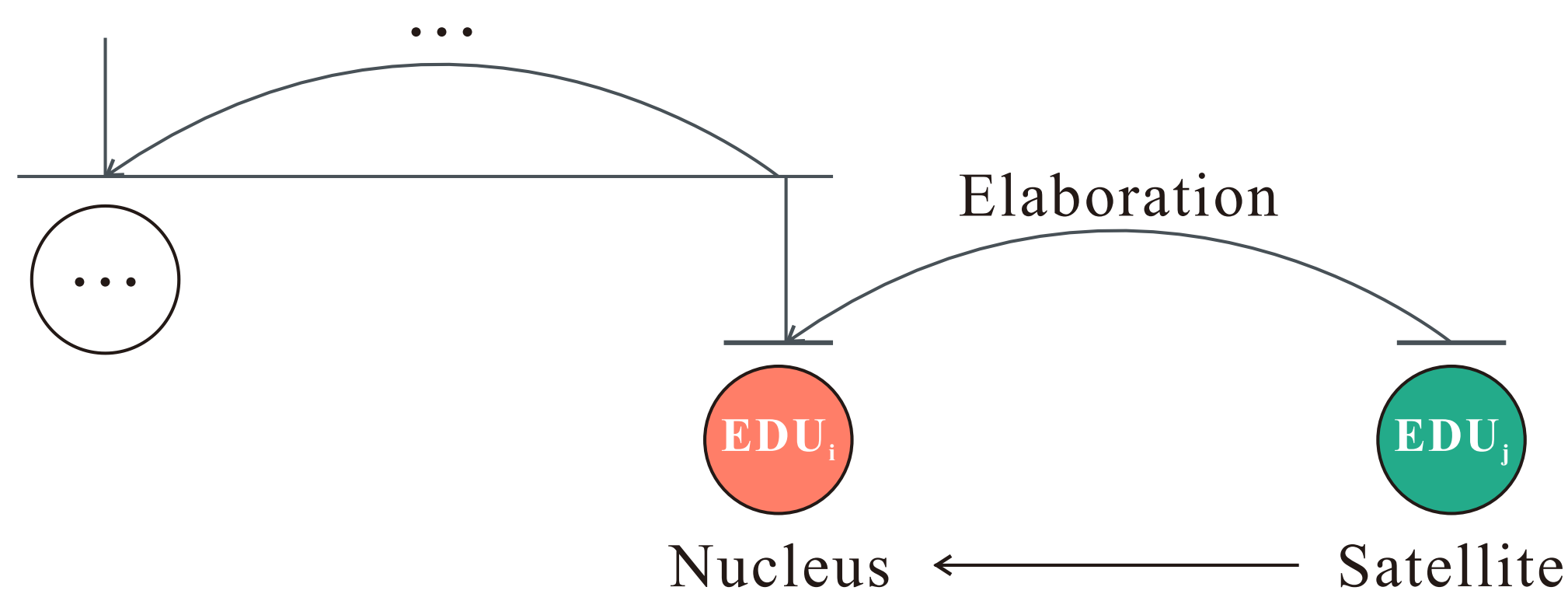


Figure 2. RST tree structure

Step 2: Plan Generation

- Use **GPT-4o** to generate questions from target sentences + context
- Questions trigger explanatory content (based on QUD framework)
- Create ordered sequence of plan questions $b = (q_1, q_2, \dots, q_n)$

Planning questions

q1: How does the cerebellum use feedback to adjust the timing of movements in a sequence?
q2: How does the cerebellum use feedback from one blink to trigger the next in a sequence?
q3: How can a blink in one eye trigger a blink in the other eye?

Processed summary

[The cerebellum utilizes ... previous actions.]^{t1} [Imagine the cerebellum ... based on what was seen.]^{e1} But how exactly does it achieve this? [To investigate, we ... to trigger the next.]^{t2} [As expected, after ... chain of movements.]^{e2} Control ... previous actions. [Building on this, we trained ... earlier blinks in the sequence.]^{t3} [We further found ... these movements.]^{e3} This raises the possibility ... should explore.

Target EDU Explanatory EDU

Figure 3. Plan generation pipeline

Method

Step 3: Two Model Variants

- Plan-Output** (End-to-End): Train $P(b, s|x)$ to generate plan (b) + summary (s) jointly based on input document (x)
- Plan-Input** (Pipeline): **PG Module** $P(b|x)$ generates plan, **SG Module** $P(s|x, b)$ generates summary

Main Results

- Plan-Input** achieves **SOTA performance** across all metrics
- More explanations**: ExpRatio 17.68% vs. 13.61% (Mistral_{FT})
- Better factual consistency**: VeriScore 0.71 vs. 0.56 (Mistral_{FT})
- Higher readability**: D-SARI 37.18 vs. 30.11 (Mistral_{FT})

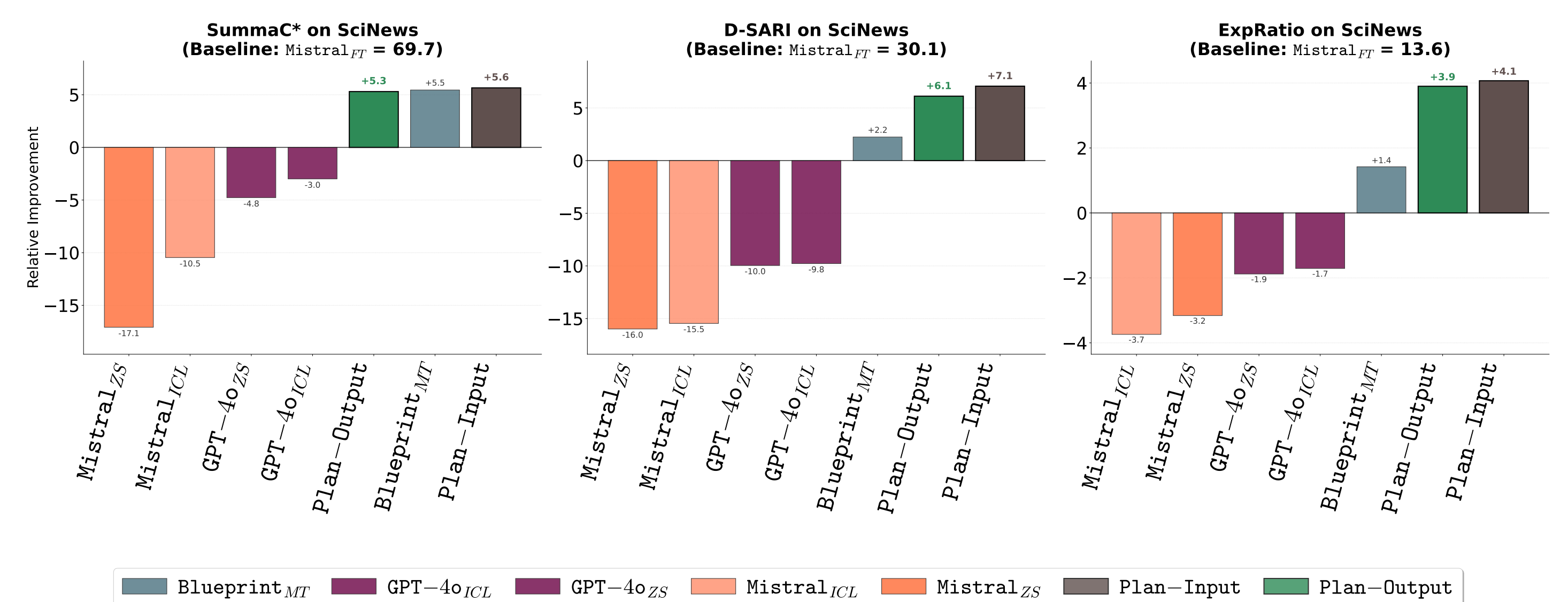


Figure 4. Model performance comparison across different metrics

Human Evaluation: Humans remain superior, but **Plan-Input** outperforms all baselines, achieving near human-level **accessibility** and substantial improvements in **explanation quality**.

Controllability of Explanatory Content

- Our models can **control explanation types** by modifying plans
- Deleting specific RST relation questions reduces the corresponding explanations in the output

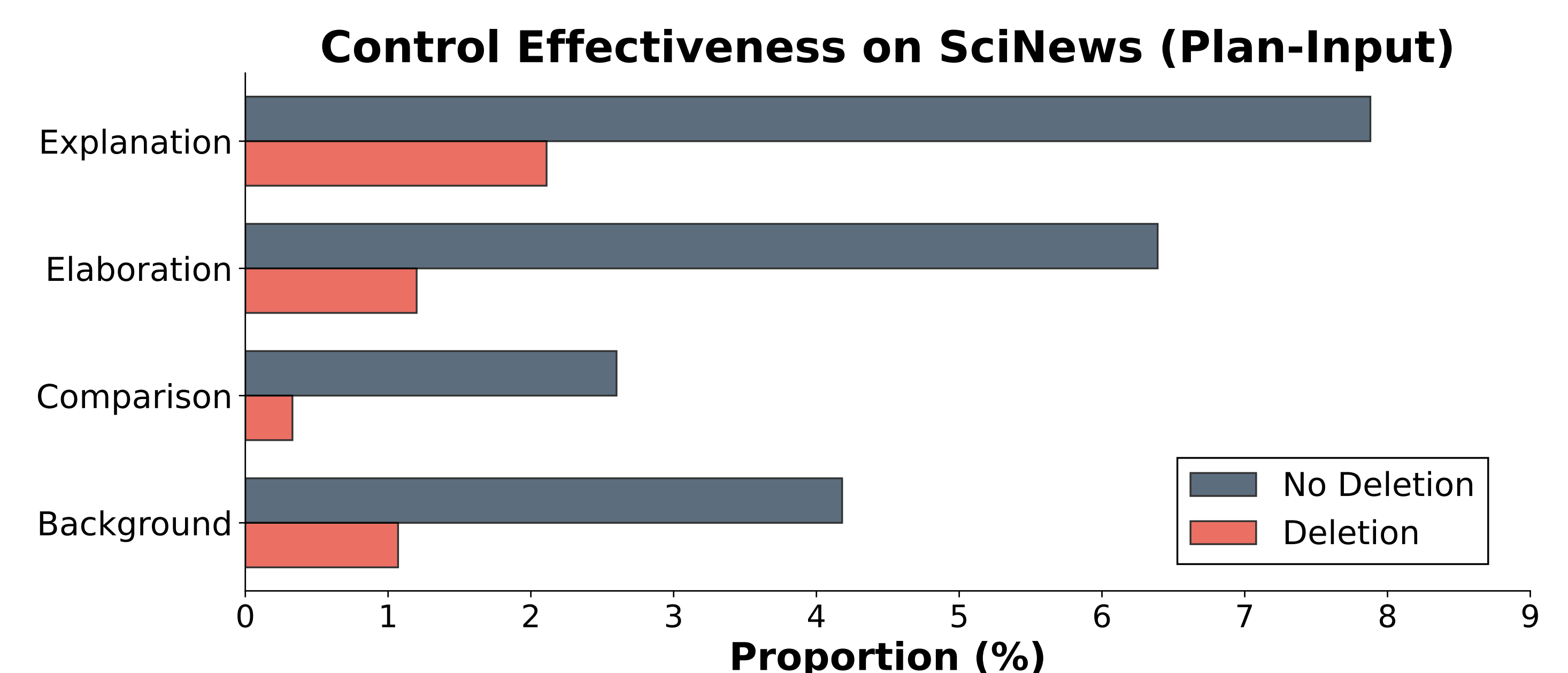
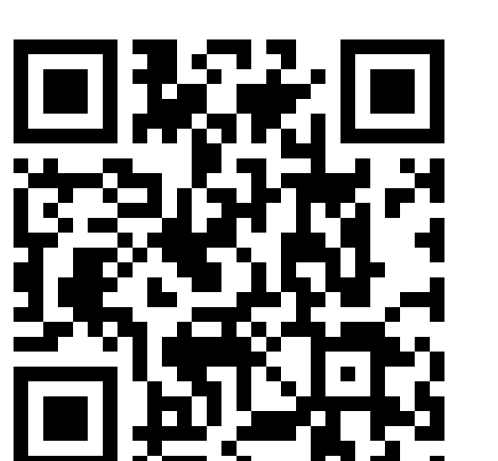


Figure 5. Control effectiveness across different explanation types

Conclusion

- Introduced **explanatory summarization** task for controlled lay summary generation
- Developed **discourse-driven planning** using RST + QUD frameworks
- Proposed two model variants: **Plan-Input** and **Plan-Output**
- Achieved **SOTA performance** on 3 datasets with significant improvements

Project Info



dongqi.me/projects/ExpSum